ValPay Portfolio 3: Transactions and Concurrency

MIT 8103 Advanced Database Systems · Evidence

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Compiled from `portfolio-3-transactions-concurrency/evidence`: 6 screenshots in `screenshots/` and script output in `script-output/` (`transaction_run.txt`). Text output is reproduced verbatim.

1. BALANCES BEFORE THE TESTS

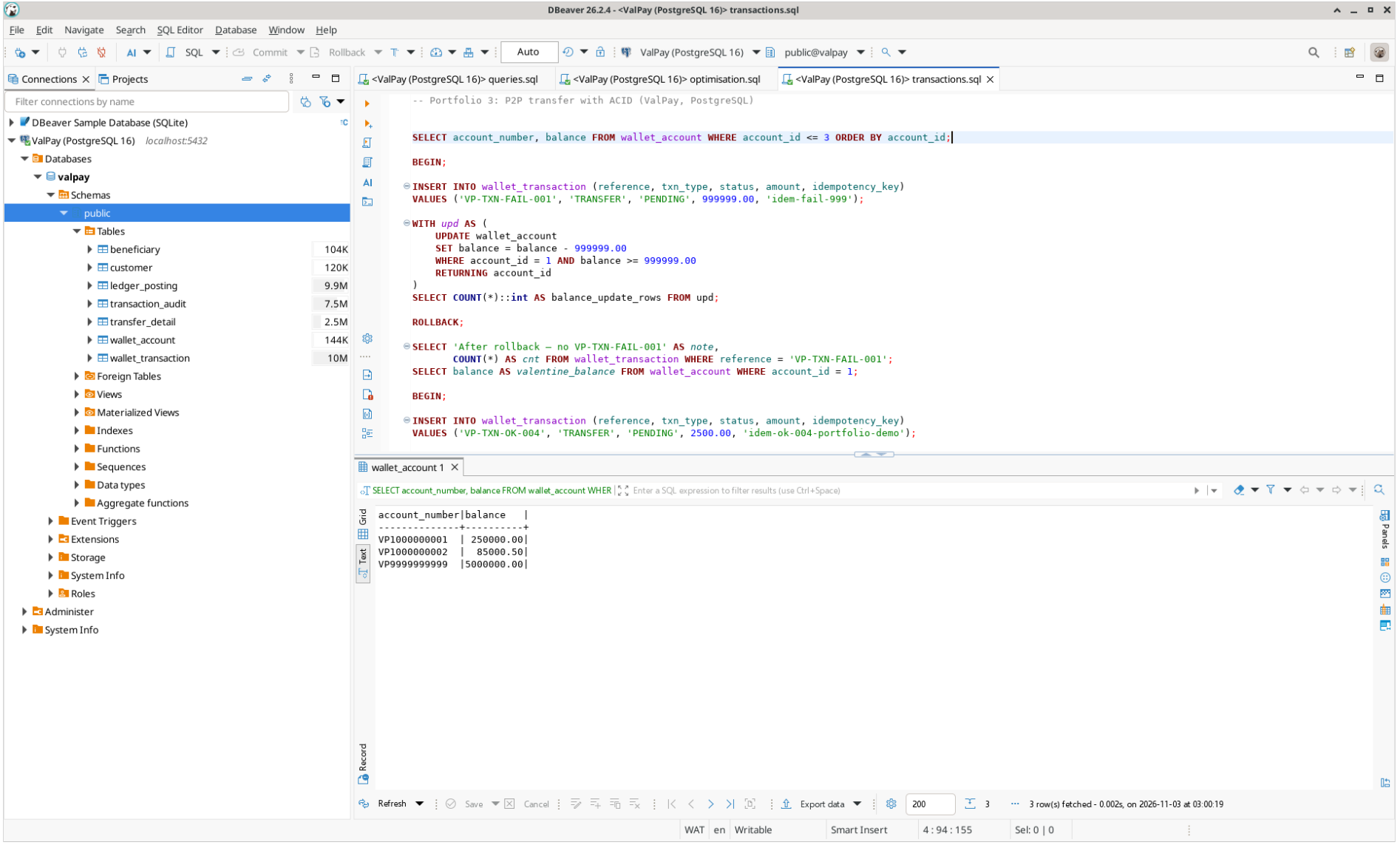


Figure 1. *Balances of accounts VP1000000001, VP1000000002 and VP9999999999 before the transfer tests.*

Source: evidence/script-output/transaction_run.txt

```
Border style is 2.
Output format is aligned.
=== Balances before transfer tests ===
+----------------+------------+
| account_number |  balance   |
+----------------+------------+
| VP1000000001   |  250000.00 |
| VP1000000002   |   85000.50 |
| VP9999999999   | 5000000.00 |
+----------------+------------+
(3 rows)
```

2. TEST A: FAILED TRANSFER AND ROLLBACK

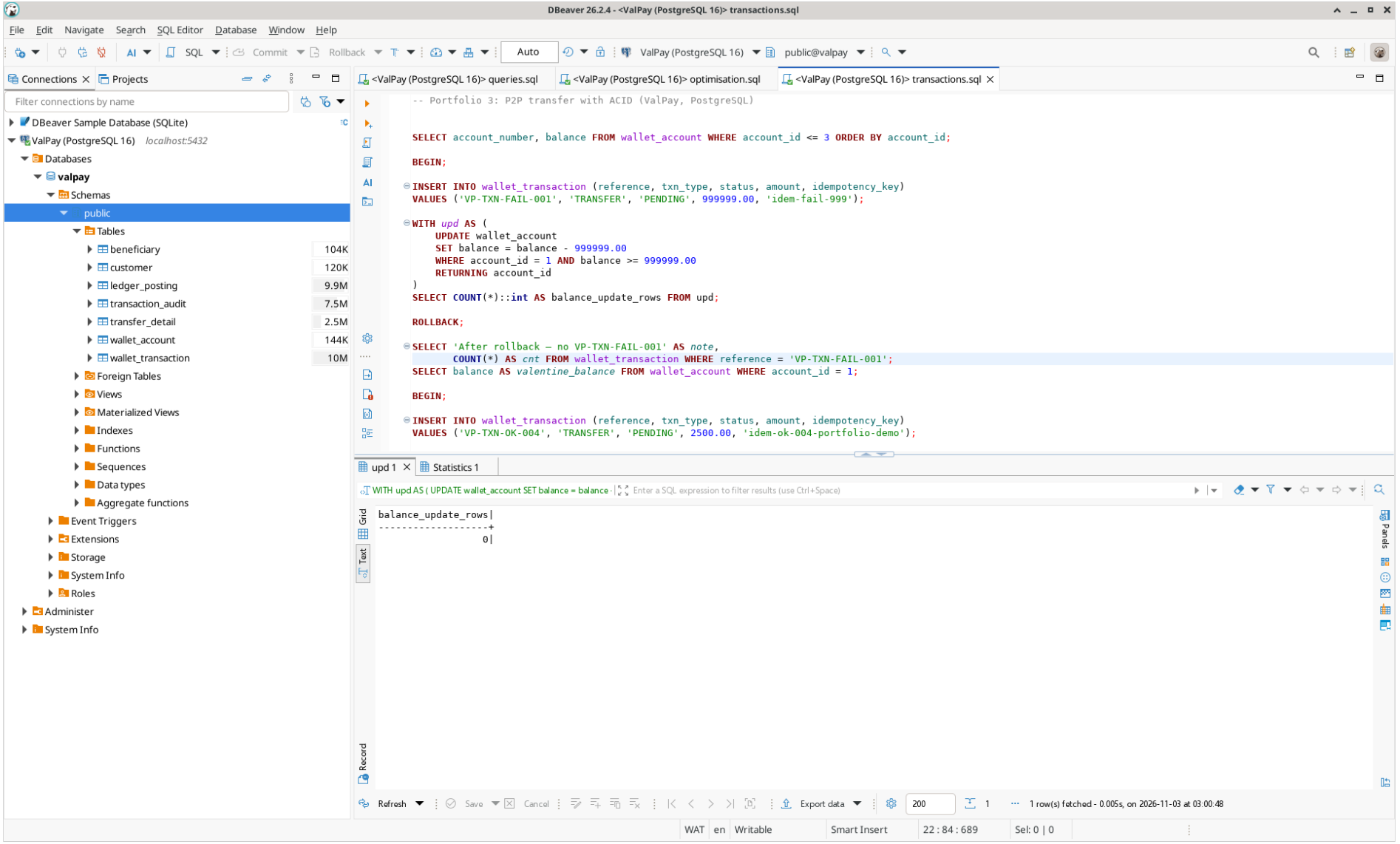


Figure 2. *Test A: the guarded balance UPDATE for 999,999.00 NGN affects 0 rows.*

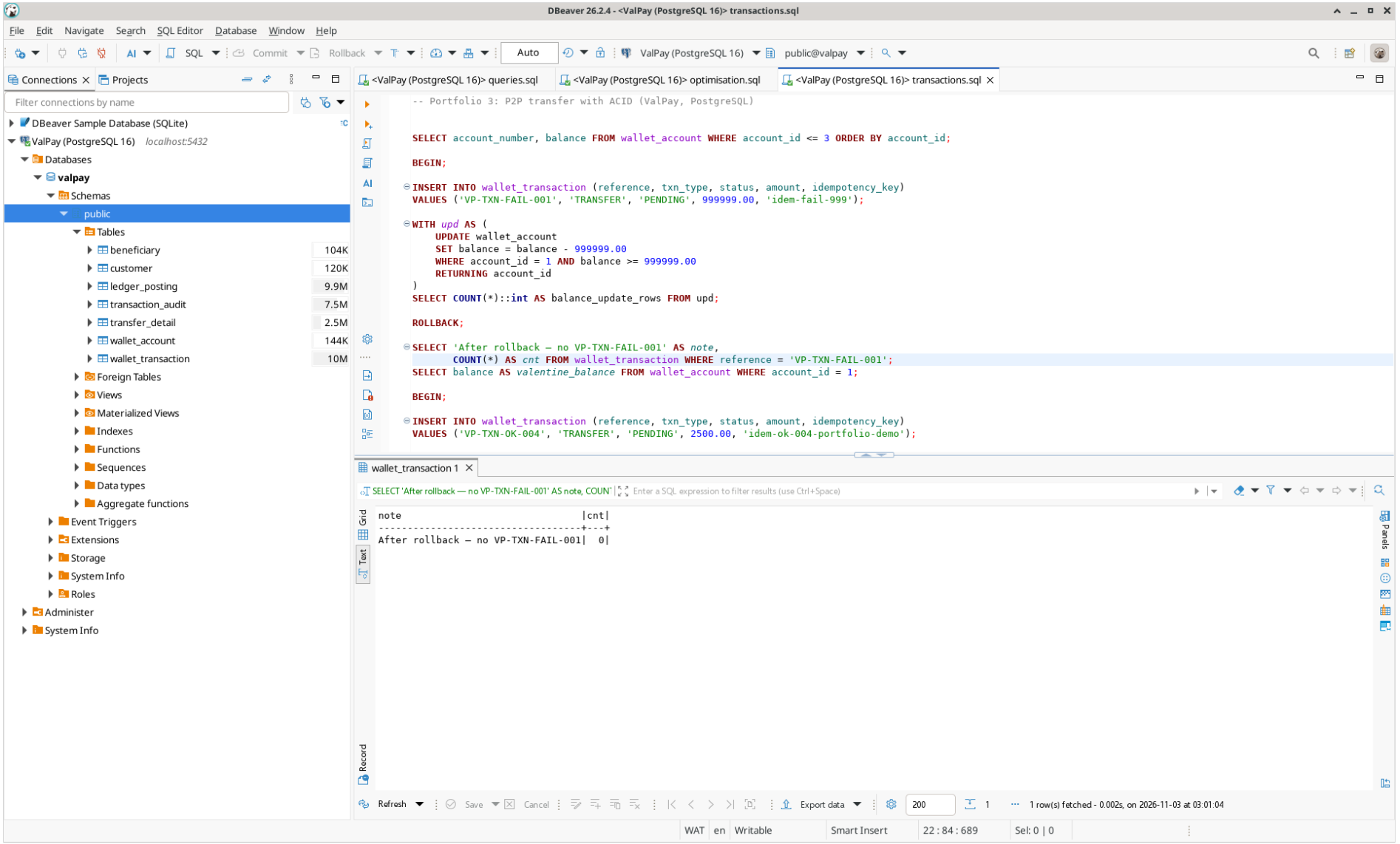


Figure 3. *Test A after ROLLBACK: no wallet_transaction row with reference VP-TXN-FAIL-001.*

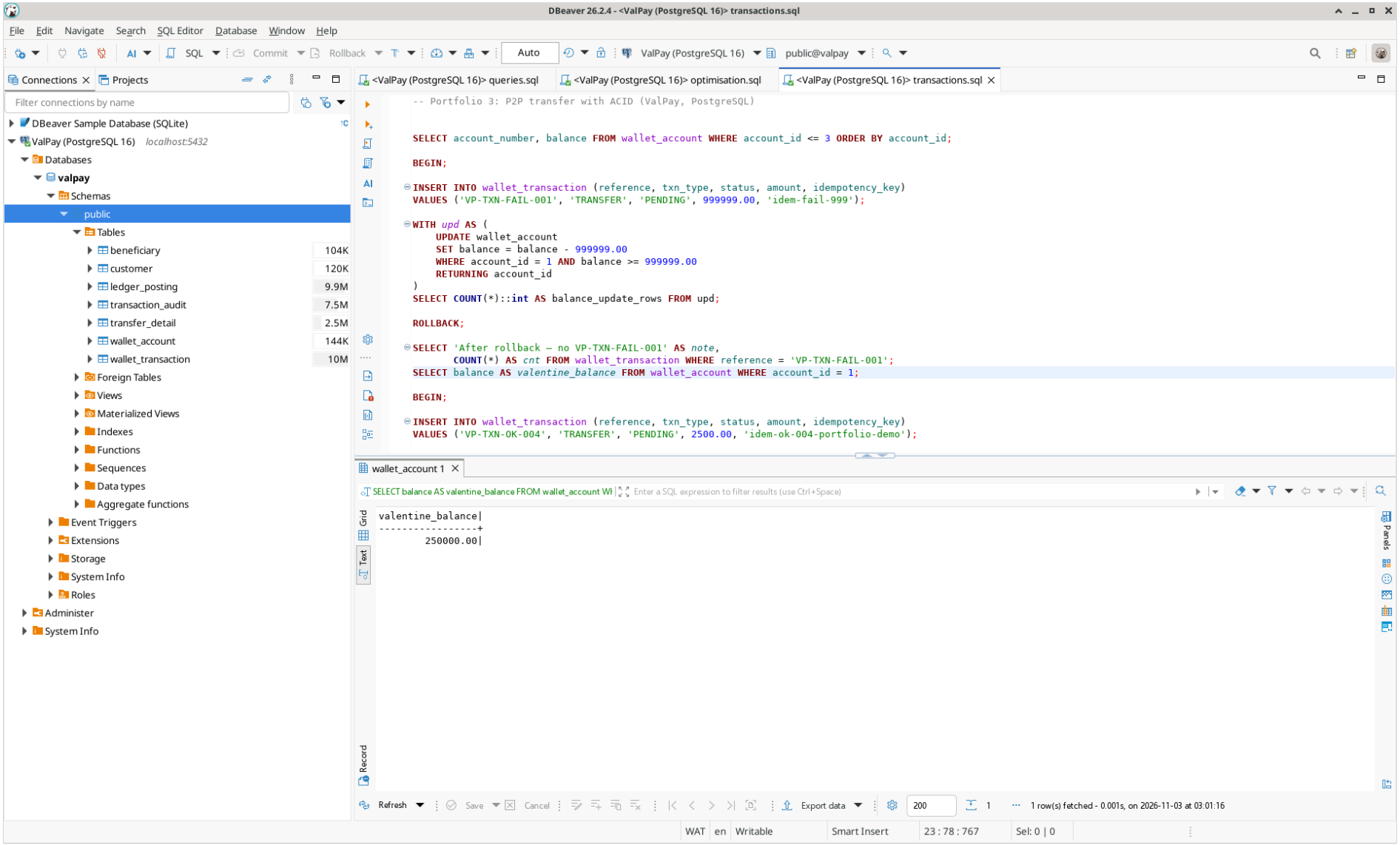


Figure 4. *Test A after ROLLBACK: balance of account 1 unchanged.*

Source: `evidence/script-output/transaction_run.txt`

```
=== Test A: FAILED transfer — insufficient funds ===
BEGIN
```

```
INSERT 0 1
+---------------------+
| balance_update_rows |
+---------------------+
|                   0 |
+---------------------+
(1 row)

ROLLBACK
+--------------------------------------+-----+
|                note                  | cnt |
+--------------------------------------+-----+
| After rollback — no VP-TXN-FAIL-001 |   0 |
+--------------------------------------+-----+
(1 row)

+-------------------+
| valentine_balance |
+-------------------+
|         250000.00 |
+-------------------+
(1 row)
```

3. TEST B: SUCCESSFUL TRANSFER AND COMMIT

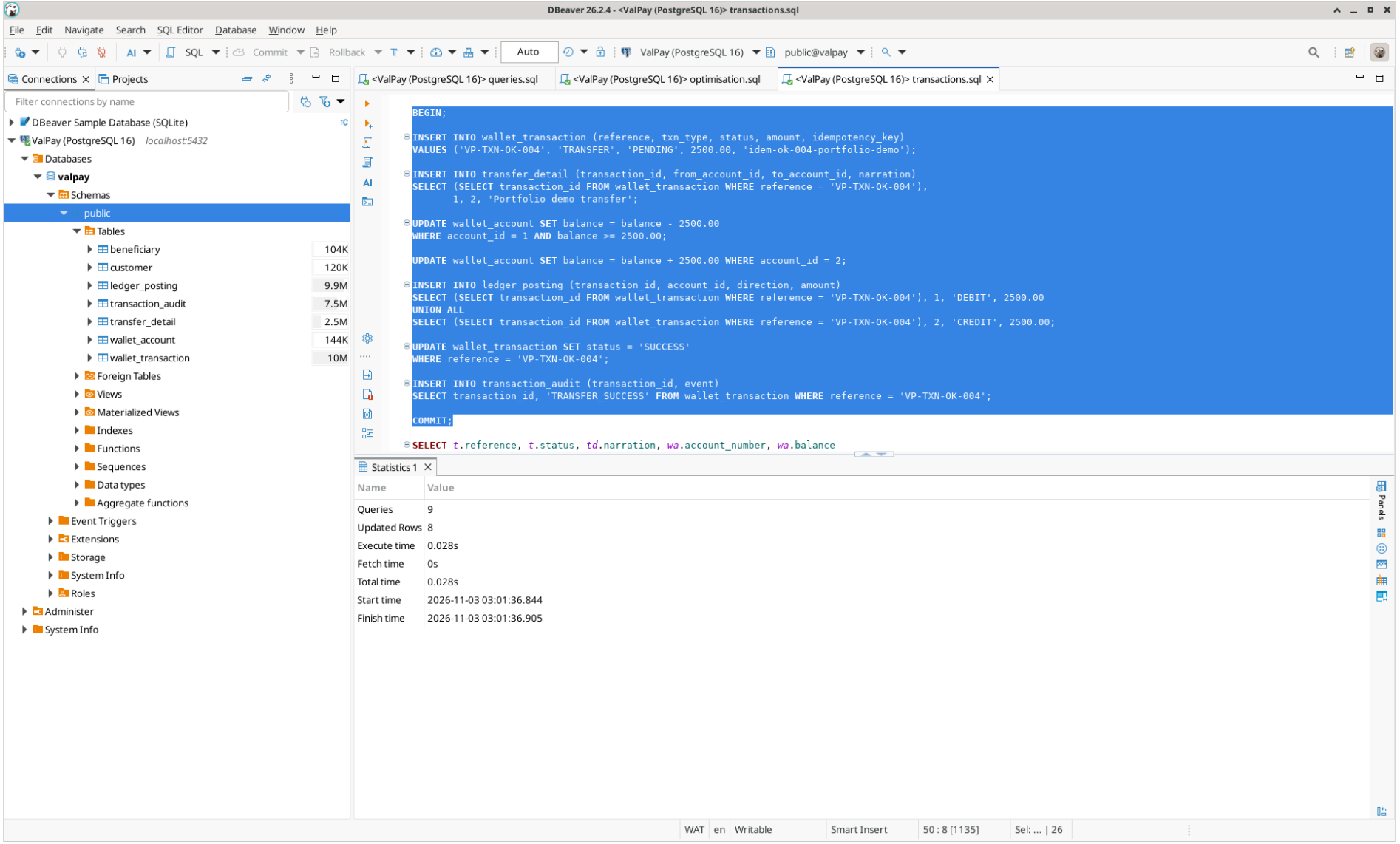


Figure 5. *Test B: the 2,500 NGN transfer script from account 1 to account 2 executed and committed.*

Source: evidence/script-output/transaction_run.txt

```
=== Test B: SUCCESSFUL P2P transfer 2,500 NGN (account 1 -> 2) ===
BEGIN
INSERT 0 1
INSERT 0 1
UPDATE 1
UPDATE 1
INSERT 0 2
UPDATE 1
INSERT 0 1
COMMIT
```

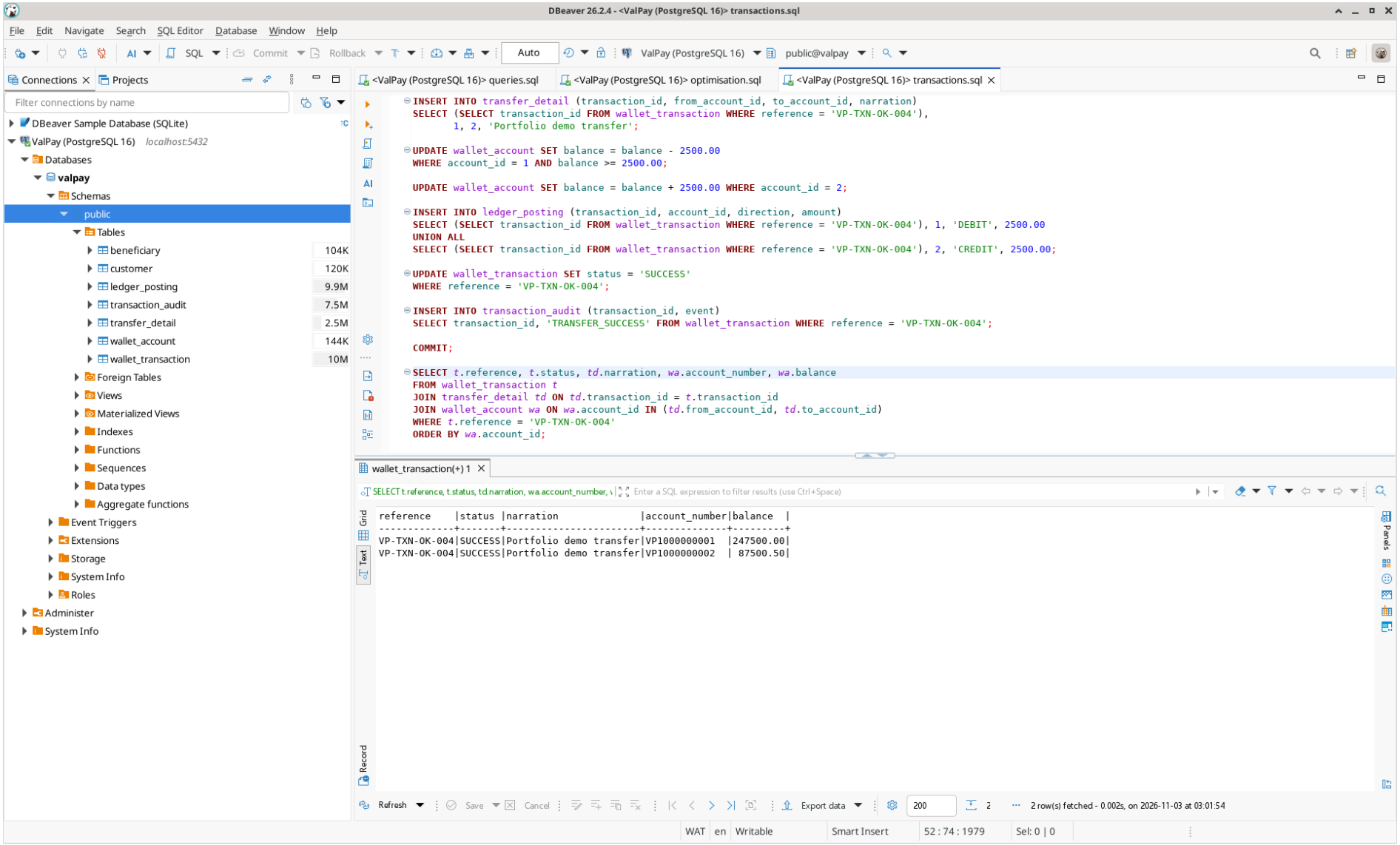


Figure 6. *After Test B: VP-TXN-OK-004 with status SUCCESS and the new balances of both accounts.*

Source: `evidence/script-output/transaction_run.txt`

```
=== After successful transfer ===
+---------------+---------+-------------------------+----------------+-----------+
```

```
|   reference   | status  |        narration        | account_number |  balance  |
+---------------+---------+-------------------------+----------------+-----------+
| VP-TXN-OK-004 | SUCCESS | Portfolio demo transfer | VP1000000001   | 247500.00 |
| VP-TXN-OK-004 | SUCCESS | Portfolio demo transfer | VP1000000002   |  87500.50 |
+---------------+---------+-------------------------+----------------+-----------+
(2 rows)
```